FORMULA STUDENT INSPECTION SHEET

ELECTRIC – Copy of FSG Inspection Sheet for Formula IHU

UNIVERSITY:
VEHICLE NUMBER:
INSPECTION ORDER:

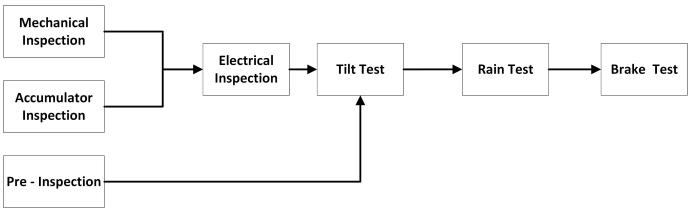
SES PASSED:
IADR PASSED:
-

ESF PASSED:
TS VOLTAGE:
BODY PROTECTION R:

Used Symbols:
- ● Information
- ▶ Action
- △ Check in responsibility of the team
- ◯ Check
- ⊙ Check optional, if Mechanical, Electrical, or Accumulator Inspection at FSA, FSCH, FSN, FSPL, or FSPT is passed

NOTES:
- This form must stay with the push bar at all times!
- Technical inspection approval voids if inspection sheet is lost.
- If there is a conflict between this form and the rules, the rules prevail.

Present the vehicle for inspection in the following order:



PART I: COMMENTS FROM DOCUMENT REVIEW

ACCUMULATOR

ELECTRICAL

MECHANICAL

PART II: PRE-INSPECTION

☐ TIS STATUS UPDATE

► Set online TIS status to *Present*

► Write down inspector names legibly, sign only when passed

☐ TIRES

1 ◯ **DRY TIRES** - Make: _____

2 ◯ **DRY TIRES** - Size: _____

3 ◯ **DRY TIRES** - Compound: _____

4 ◯ **RAIN TIRES** - Make: _____

5 ◯ **RAIN TIRES** - Size: _____

6 ◯ **RAIN TIRES** - Compound: _____

7 ◯ **RAIN TIRES** - 2,4 mm min. tread depth molded by tire manufacturer

☐ DRIVER GEAR & SAFETY

8 ◯ **FIRE EXTINGUISHERS** - Two (2) hand-held, foam type (10BC, 1A10BC, 34B, 5A 34B, 20BE or 1A 10BE), with pressure/charge gauge, presented at inspection.).

9 ◯ **UNDERWEAR** - Nomex or equivalent, fire resistant underwear (no cotton, no polyester, no bare skin). No holes.

10 ◯ **SOCKS** - Nomex or equivalent, fire resistant socks (no cotton, no polyester, no bare skin). No holes.

11 ◯ **GLOVES** - Fire resistant material. Leather allowed only over fire resistant material. FIA hologram present. No holes.

12 ◯ **ARM RESTRAINTS** - SFI Standard 3.3 or equivalent.

13 ◯ **HELMETS** - Snell SA2020, SA2025, EA2016 or newer. 31.1/2015, 31.1/2020 or newer. FIA 8860-2010, FIA 8860-2018, FIA 8859-2015, FIA 8859-2024 or newer. Closed face, no open face, must have integrated shield (no dirtbike helmets). No camera mounts. Communication equipment only allowed, if compliant with FIA regulations (International Sporting Code - Appendix L).

14 ◯ **FHR/HANS** - If used, must be certified to one of these standards: FIA 8858-2010, FIA 8860-2004, SFI 38.1.

15 ◯ **DRIVER SUITS** - Single piece SFI 3.2A/5 (or higher), SFI 3.4/5 (or higher), FIA 8856-2000/2018 (or higher), and LABELED AS SUCH. FIA hologram present. No holes.

16 ◯ **SEWING OR STITCHING** - Teams must show compliance to T13.3 if driver's clothing is embroidered. Fire resistant material must be used, examples: Nomex, Aramid, Belcotex and Indura.

17 ◯ **HAIR COVER** - Fire resistant (Nomex or equiv.) balaclava of full helmet skirt REQUIRED FOR ALL DRIVERS. No holes.

18 ◯ **SHOES** - SFI 3.3 or FIA 8856-2000/2018. No holes.

19 △ **COMMUNICATION EQUIPMENT** - Must be securely attached to the vehicle. Must not be worn on the driver, except for headphones and microphone. Must be worn during egress test, if used at any discipline.

☐ TIS STATUS UPDATE

► Set online TIS status to *Passed* or *Failed*

NON-COMPLIANCE / COMMENTS

Item #	Non-compliance / Comment	Inspector Name

APPROVAL

	Inspector Names	Date, Time	Signatures when passed
1.	_____ / _____	_____	_____

PART III: EGRESS TEST

☐ DRIVER POSITION

20 ◯ **ARM RESTRAINTS**- Must be installed so the driver can release them and exit unassisted regardless of vehicle's position.

21 ◯ **HEAD RESTRAINT**- Near vertical. Max. 25 mm from helmet. Helmet contact point 50 mm min. from any edge.

22 ◯ **MAIN HOOP & FRONT HOOP HEIGHTS** - Helmet of driver to be 50 mm below line between top of front and main roll hoop AND between top of main hoop to rear attachment point of main hoop bracing.

23 ◯ **SHOULDER HARNESS MOUNTING** - Angle from shoulder between 10 deg. up and 20 deg. down to horizontal.

☐ DRIVER EGRESS TEST

- All drivers must be able to exit the vehicle in less than 5s.
- Driver must be seated in ready to race condition.
- Communication equipment installed, if intended to use at any dynamic discipline.

☐ EGRESS PROCEDURE

- ► Both hands on the steering wheel. (in all possible steering positions)
- ► Pressing cockpit-mounted shutdown button.
- The egress time will stop when the driver has both feet on the ground

DRIVER APPROVAL & RUN DOCUMENTATION

	Driver's Name	Wristband ID	Signature Inspector - when passed	Comm. equip.?	Acc	Skid Pad	AutoX	Endu-rance
1.				☐			☐☐	☐
2.				☐			☐☐	☐
3.				☐			☐☐	☐
4.				☐			☐☐	☐

Checked by officials only after a dynamic run!

PART IV: ACCUMULATOR INSPECTION

The time limit for this part of the inspection is 105 minutes. Continuation of the inspection is possible after requeueing.
During technical inspection all work carried out on the accumulator must be approved by a technical inspector.

INSPECTOR NAMES

_____ / _____

_____ / _____

☐ TIS STATUS UPDATE

► Set online TIS status to *Present*

☐ COMMENTS

► Check comments from first page

☐ REQUIRED RESSOURCES

24 ◯ An ESO must attend.

- All accumulator containers to be used during the event.

25 ◯ No spare TS accumulator containers.

- Accumulator Container Hand Cart.
- Charger.
- Tools needed for (dis-)assembly of Accumulator Container.
- PDF or print-out of rule questions, if necessary.
- Pictures of accumulator internals, if necessary.
- Datasheets for used wiring, insulation materials, and TS components. (printed or properly sorted on one laptop, not on a cell phone)
- Samples of all wire types used inside the accumulator container.
- Samples of all used accumulator container material.
- Fully assembled spare boards of all inaccessible TS boards inside the accumulator
- Laptop and cables to display data of the AMS

☐ SAFETY BRIEFING

- no jewellery, no rings
- no cell phone
- no batch / no necklace
- no sources of distraction
- do not wear synthetic clothes
- wear safety glasses
- wear safety gloves

☐ BASIC SET OF HV-PROOF TOOLS

26 ◯ Insulated cable shear.

27 ◯ Insulated screw driver.

28 ◯ Insulated spanners (n/a if no screwed connections in TS).

29 ◯ Multimeter with protected probe tips

30 ◯ two $4\,\mathrm{mm}$ banana plug test leads ($600\,\mathrm{V}$ CAT III)

☐ SAFETY EQUIPMENT

31 ◯ Face shield.

32 ◯ Safety glasses (minimum three).

33 ◯ HV insulating gloves (minimum two pairs).

34 ◯ HV insulating blankets (two) (min $1\,\mathrm{m}^2$) with label or serial number and datasheet.

☐ SELF DEVELOPED PCBS

► Ask for fully assembled spare PCB of self developed PCBs inside accumulator container.

35 ◯ Sufficient spacing regarding system voltage and implementation, mind new rules.

36 ◯ Sufficient insulation and temperature rating of coating if used, check datasheets.

37 ◯ Coating process according to datasheet

☐ HAND CART

38 ◯ Hand cart present with four wheels. Max. dimensions $1200\,\mathrm{mm}$ x $800\,\mathrm{mm}$.

39 ◯ Hand cart has always on type brake system.

40 ◯ Hand cart can be easily moved if brake is released.

41 ◯ The accumulator must be mechanically fixed to the handcart while on the handcart.

42 ◯ The accumulator must be protected from vibrations and shocks.

43 ◯ Firewall (same width as hand cart, from lowest point to $30\,\mathrm{cm}$ above TSAC/handle) must protect operator.

44 ◯ Label according to EV5.3.8 on hand cart firewall, max $1.3\,\mathrm{m}$ above ground.

☐ ACCUMULATOR CONTAINER LABELING

45 ◯ Vehicle number, university name and ESO phone number(s) written on a high contrast background.

46 △ Roman Sans-Serif characters of at least $20\,\mathrm{mm}$high are used.

47 ◯ Warning stickers with side length of $\geq 100\,\mathrm{mm}$ and text "Always Energized" and "High Voltage" (if TS $>60\,\mathrm{V}$) installed. (triangle with black lightning bolt on yellow background)

☐ CHARGER ASSEMBLY

48 ◯ Completely closed. Check opening in HV/TS enclosures, try to reach HV/TS potentials with insulated test probe ($100\,\mathrm{mm}$ length, $6\,\mathrm{mm}$ diameter).

49 ◯ Interlock integrated.

50 ◯ TSMP integrated

51 ◯ Emergency shutdown button integrated.

52 △ Emergency shutdown button $\geq 24\,\mathrm{mm}$ diameter.

53 ◯ TSAL green light indicator integrated. Must be hard-wired electronics.

54 ◯ TS wiring is orange, marked with gauge, temperature rating $>85\,^{\circ}\mathrm{C}$ and voltage rating.

55 ◯ Conductive parts of charging equipment and accumulator are connected to protective earth (PE) while charging. Mind new grounding rules, see EV 3.1

56 ◯ Switches, plugs and indicators must be labeled.

☐ DIS-CHARGE CIRCUIT AND BODY PROTECTION RESISTORS

► Switch off Charger. Measure resistance between TS+ and TS- measuring points.

57 ◯ Resistance is $30\,\mathrm{k\Omega}$ [1] + discharge resistor

58 ◯ Body protection resistor power rating is $>6.1\,\mathrm{W}$ [2]

[1] 2 x Body Protection Resistor (BPR)

[2] sufficient to short circuit TS+ and TS-

☐ OPEN ACCUMULATOR CONTAINER

- ► Open container housing, remove maintenance plugs.
- ► Check if no voltage is present.
- ► Get mechanical inspector for ASES checks.

☐ ASSEMBLY

59 ◯ All components and parts of the accumulator container need to be properly fixed.
60 ◯ TS potentials are insulated against inner wall of accumulator container if container made from conductive material.
61 ◯ A minimum of $30\,\%$ of cells are monitored with temperature sensors.
62 ◯ Every temperature sensor placed on negativ terminal of monitored cell or in $<10\,\mathrm{mm}$ distance on busbar.
63 ◯ No cells are damaged or can be damaged by the segment structures.
64 ◯ No soldering in high current path
65 ◯ Every container contains at least one appropriately sized and rated fuse.
66 ◯ Every container contains at least two appropriately sized and rated isolation relays (current and voltage).
67 ◯ Isolation relays and fuses are separated from cells by barrier according UL94-V0 or equivalent.
68 ◯ Pre-charge relay is of mechanical type with appropriate voltage rating.
69 ◯ Maintenance plugs are located at both poles of each stack (including first and last stack).
70 ◯ Maintenance plugs removable without tools.
71 ◯ Maintenance plugs have positive locking mechanism.
72 ◯ Maintenance plugs must not be able to unintentionally create circuits or short circuits.
73 ◯ Stacks separated by Maintenance plugs $\leq 120\,\mathrm{V\,DC}$.
74 ◯ Stacks separated by Maintenance plugs $\leq 6\,\mathrm{MJ}$.
75 ◯ Stacks are insulated and separated by a fire resistant barrier according to UL94-V0 for min. used thickness or equivalent.
- ► Check opening in TS enclosures, try to reach TS potentials with insulated test probe ($100\,\mathrm{mm}$ length, $6\,\mathrm{mm}$ diameter).

76 ◯ If fully closed, equalizing valve implemented.

☐ WIRING

77 ◯ All TS wires have proper overcurrent protection.
78 ◯ No other wires than TS wires are orange.
79 ◯ Securely anchored to withstand at least $200\,\mathrm{N}$, if outside of enclosure.
80 ◯ Located out of the way of possible snagging or damage.
81 ◯ TS and LV wires separated (not valid for Interlock).
82 ◯ Every wire used in the Accumulator container (TS and LV) is rated for $\geq 604.8\,\mathrm{V}$ [3].
83 ◯ Possible to clearly assign and prove gauge, temperature and voltage rating of TS wires.
84 ◯ Positive locking mechanism or if no positive locking possible, automotive certified components.
- ► Check if insulated tools needed for the assembly of certified components are available

85 ◯ Insulation is not only insulating tape or rubber-like paint.
86 ◯ IMD is integrated into the charging system
87 ◯ One IMD ground line is connected to the accumulator container and one ground line is connected to the charger casing by a separate wired connection

☐ ACCUMULATOR MANAGEMENT SYSTEM

- ► Disconnect AMS current sensor connector

88 ◯ The AMS must open the shutdown circuit within $0.5\,\mathrm{s}$.
- ► Disconnect any other AMS internal connector

89 ◯ The AMS must open the shutdown circuit within $1\,\mathrm{s}$.
- ► Disconnect one SINGLE voltage sense wire, if any wires used.

90 ◯ The AMS must open the shutdown circuit within $0.5\,\mathrm{s}$.
- ► Disconnect one SINGLE temperature sense wire, if any wires used.

91 ◯ The AMS must open the shutdown circuit within $1\,\mathrm{s}$.
- ► Ask the team to connect their laptop to the AMS.
- ► Connect charger to battery/batteries, start charging process.

92 ◯ Cell voltages can be displayed.
93 ◯ Cell temperatures can be displayed.
94 ◯ Plausible accumulator current can be displayed.

☐ INDICATOR LIGHT OR VOLTMETER AND TSAL GREEN LIGHT INDICATOR

95 ◯ Red indicator light or voltmeter installed
96 ◯ Marked with “Voltage Indicator”
97 ◯ Visible while opening the battery connector.
98 ◯ Hard wired electronics, supplied by TS
- ► Activate LVS

99 ◯ TSAL green light indicator on.
100 ◯ Green and easily visible.
- ► Connect power supply with $60\,\mathrm{V\,DC}$ [4] to accumulator TS connector. Use proper plugs, no measuring probes.

101 ◯ Indicator light on or voltmeter showing present TS voltage.
102 ◯ Indicator light continuously on with same brightness.
103 ◯ Visible in bright sunlight.
104 ◯ TSAL green light indicator off.

[3] max. TS voltage

[4] $60\,\mathrm{V}$ or half the nominal tractive system voltage, whichever is lower

☐ CLOSE ACCUMULATOR CONTAINER

105 ◯ All ASES checks have been performed
- ▶ Install maintenance plugs, close container housing

106 ◯ Seal accumulator container

☐ INSULATION MEASUREMENT TEST

- ▶ Check low resistance connection between LV ground MP and PE/casing
- ▶ Choose test voltage to $500\,\mathrm{V}$. [5]
- ▶ Connect insulation tester to charger TS+ and LV ground.
- ▶ Connect charger (do not activate charger) to accumulator, keep AIRs opened.
- ▶ Measure resistance: $R_{\mathrm{iso}+}$ = MΩ

107 ◯ Resistance is much higher than $315\,\mathrm{k}\Omega$ [6].
- ▶ Connect insulation tester to TS- and LV ground.
- ▶ Measure resistance: $R_{\mathrm{iso}-}$ = MΩ

108 ◯ Resistance is much higher than $315\,\mathrm{k}\Omega$ [6].

109 ◯ Resistances are nearly equal.

☐ CHARGER SHUTDOWN CIRCUIT

- ▶ Connect charger to battery/batteries, start charging process

110 ◯ Voltage indicator shows that HV is present
- ▶ Press shutdown button

111 ◯ AIRs open

112 ◯ Voltage indicator shows voltage <$60\,\mathrm{V}$
- ▶ Start charging, unplug TS accumulator connector

113 ◯ AIRs open.

114 ◯ Charger disabled, no voltage at charger connector
- ▶ Reconnect TS accumulator connector

115 ◯ Charger stays off, no voltage at TSMPs

☐ INSULATION MONITORING DEVICE

- ▶ R_{Test} = $120\,\mathrm{k}\Omega$ [7]
- ▶ Activate charger output, connect R_{Test} between TS+ and LV GND.

116 ◯ Shutdown circuits opens within $30\,\mathrm{s}$.

117 ◯ TS voltage decreases below $60\,\mathrm{V\,DC}$ within $5\,\mathrm{s}$ after shutdown circuit opens.

118 ◯ Reactivation of charger output is not possible.
- ▶ Push the reset button, if any.

119 ◯ Reactivation of charger output is not possible.
- ▶ Remove R_{Test}. Wait $40\,\mathrm{s}$ until IMD resets status output.

120 ◯ Reactivation of charger output is not possible.
- ▶ Activate TS, connect R_{Test} between TS- and LV GND.

121 ◯ Shutdown circuits opens within $30\,\mathrm{s}$.

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[5] $U_{max} \leq 250\,V_{DC}$ U_{Test} = $250\,V_{DC}$ | $U_{max} > 250\,V_{DC}$ U_{Test} = $500 V_{DC}$

[6] Minimal Resistance = $500\,\Omega/\mathrm{V} \cdot V_{\mathrm{max}}$ + BPR

[7] R_{Test} = (max. TS voltage $\cdot\, 250\,\Omega/\mathrm{V}$) - BPR

☐ ASES - ACCUMULATOR CONTAINER

- ► Team must show approved ASES for accumulator container.
- ► Team must show ASES test samples for accumulator container if alternative materials are used.

122 ◯ Accumulator container manufactured according to ASES. Appropriate material and bonding quality.

123 ◯ Internal vertical walls have to be rigidly fastened to the container. Extend to the lid.

124 ◯ Holes in container only for wiring harness, ventilation, cooling or fasteners, if mechanical properties are not influenced and max $25\,\%$. AIRs and main fuse separated from cells, no holes in separation.

125 ◯ Cells securely fastened towards all 3 directions. Minimal play only for form-fitting solutions.

126 ◯ Tabs of pouch cells must not carry mechanical loads.

127 ◯ All parts carrying cells and loads: UL94-V0 certified materials.

128 ◯ External openings not pointing towards driver or hand cart operator.

129 ◯ Check if all parts and the cover/lid of the housing are rigidly fastened.

130 ◯ All used fasteners must be secured by the use of positive locking, except non-conductive and non-structural fasteners.

Item #	ASES Non-compliance / Comment	Inspector Name

	Inspector Names	Date, Time	Signatures when passed
1.	_____/ _____	_____	_____
2.	_____/ _____	_____	_____

☐ TIS STATUS UPDATE

- ► Set online TIS status to *Passed* or *Failed*

NON-COMPLIANCE / COMMENTS

Item #	Non-compliance / Comment	Inspector Name

APPROVAL

	Inspector Names	Date, Time	Signatures when passed
1.	_____/ _____	_____	_____
2.	_____/ _____	_____	_____
3.	_____/ _____	_____	_____
4.	_____/ _____	_____	_____
5.	_____/ _____	_____	_____
6.	_____/ _____	_____	_____
7.	_____/ _____	_____	_____

PART V: ELECTRICAL INSPECTION

The time limit for this part of the inspection is 105 minutes. Continuation of the inspection is possible after requeueing.
During technical inspection all work carried out on the vehicle must be approved by a technical inspector.

INSPECTOR NAMES

_____ / _____

☐ TIS STATUS UPDATE

▶ Set online TIS status to *Present*

☐ COMMENTS

▶ Check comments from first page

☐ REQUIRED RESSOURCES

131 ◯ An ESO must attend
- LV battery or cell datasheet
- For self-developed LV battery packs: an opened battery pack, laptop, and cables to display data of the AMS
- Laptop and cables to display data of the TS accumulator AMS
- Datasheets for used wiring, insulation materials, and TS components. (printed or properly sorted on one laptop, not on a cell phone)
- At least all non-passed parts of the ESF. (printed or properly sorted on one laptop, not on a cell phone)
- Samples of all wire types used for the tractive system
- Fully assembled spare boards of all inaccessible TS boards outside the accumulator
- The connector to safely close the SDC while the HVD is removed
- The connector to safely supply the TS using shrouded receptacles when the TS accumulator is unconnected
- Photographs of all inaccessible TS connections
- "TSAL green" sign

☐ TS VOLTAGE

▶ Measure voltage at TS measuring points

132 ◯ Equal or less than $60\,\mathrm{V\,DC}$

☐ LV BATTERY

133 ⊙ Voltage $\leq 60\,\mathrm{V\,DC}$
134 ⊙ Rigid and sturdy casing
135 ⊙ Li-Ion: UL94-V0 or equivalent casing
136 ⊙ Short circuit protection (e.g. fused)
137 ⊙ Li-Ion: Over current protection
138 ⊙ Proper insulation of internal electrical connections
139 ⊙ Proper mounting of cells
140 ⊙ Li-Ion: Temperature sensors for at least $30\,\%$ of the cells

☐ SELF DEVELOPED PCBS

▶ Fully assembled spare PCB of self-developed PCBs
141 ⊙ Body protection resistor power rating is $>6.1\,\mathrm{W}$ [8]
142 ⊙ Dis-charge power rating is sufficient for continuous dis-charge
143 ⊙ Sufficient spacing regarding system voltage and implementation, mind new rules
144 ⊙ Sufficient rating of isolators regarding system voltage, check datasheets
145 ⊙ Sufficient insulation and temperature rating of coating if used, datasheet available
146 ⊙ Coating process according to datasheet
147 ⊙ BSPD PCBs are standalone with only minimum interface
148 ⊙ BSPD PCBs are directly supplied from the LVMS
149 ⊙ Ends of a BSPD current transducer's auxiliary winding must be insulated

☐ MASTER SWITCHES

150 ◯ TSMS & LVMS installed easily accessible on the right side of the vehicle and located next to each other
151 △ All master switches are located above $80\,\%$ of shoulder height of percy
152 ⊙ Rigidly mounted and no need to be removed during maintenance
153 ⊙ Rotary type with removable handle
154 △ Handle length $\geq 50\,\mathrm{mm}$
155 ⊙ "ON" position in horizontal
156 ⊙ "ON" and "OFF" positions marked
157 ⊙ TSMS with locking mechanism for "OFF" position
158 ⊙ LVMS marked with "LV" and a symbol showing a red spark in a white-edged blue triangle
159 ⊙ LVMS mounted on a red circular area on high contrast background
160 △ Circular area diameter $\geq 50\,\mathrm{mm}$
161 ⊙ TSMS marked with "TS" and triangle with black lightning bolt on yellow background
162 ⊙ TSMS mounted on an orange circular area on high contrast background
163 △ Circular area diameter $\geq 50\,\mathrm{mm}$

☐ MEASURING POINTS

164 ◯ Two TS measuring points on exclusive orange background
165 ◯ A black LVS ground measuring point installed
- An additional LVS ground measuring point may be installed for 4-wired grounding checks

166 ⊙ Next to the master switches
167 ⊙ $4\,\mathrm{mm}$ shrouded banana jacks

[8] sufficient to short circuit TS+ and TS-

168 ⊙ Non conductive cover
169 ⊙ Cover removable without tools
170 ⊙ Correctly marked (“TS+”, “TS-”, “GND”)

☐ TS SHUTDOWN DEVICES

171 ◯ Two red latching shutdown buttons installed next to the main hoop, right and left on the vehicle at approx. height of the driver’s head
172 ⊙ Marked with red sparked sticker
173 △ Diameter $\geq 39\,\mathrm{mm}$
174 ◯ One red latching shutdown button installed inside the cockpit
175 ⊙ Marked with red sparked sticker
176 ⊙ Easy actuation by the driver
177 △ Diameter $\geq 24\,\mathrm{mm}$
178 ◯ Inertia switch upright and rigidly mounted to the chassis and can be demounted for functionality test

☐ HIGH VOLTAGE DISCONNECT

179 ◯ Clearly marked with "HVD"
180 △ Distance to ground greater than $350\,\mathrm{mm}$
181 ⊙ No remote actuation (e.g. through wires)
182 ⊙ Integrated interlock
► Let ESO remove the HVD
183 ⊙ Removed within $10\,\mathrm{s}$ without tools
184 ◯ TS protection still given (insulated test probe). If a dummy connector is used, it must be stored at the push bar.

☐ ACCELERATOR PEDAL POSITION SENSOR

185 ◯ Returns to the original position if not actuated
186 ⊙ At least two sensors with different, non-intersecting transfer functions, with either different gradients and/or offsets to the others are installed. (For digital sensors, a checksum is necessary)
187 ⊙ Sensors are protected from being mechanically overstressed (positive stop of the pedal)
188 ⊙ Minimum two springs installed to return pedal
189 ⊙ Each spring still returns pedal with the second one disconnected (springs in the torque encoders not counted)

☐ BRAKE LIGHT

190 ◯ Only one brake light in red color
191 ⊙ Located on vehicle centerline, height between wheel centerline and drivers shoulder
192 ⊙ Round, triangle, or rectangular on black background
193 △ $15\,\mathrm{cm}^2$ minimum illuminated area *OR* LED strips with a total length greater than $150\,\mathrm{mm}$ with elements $<20\,\mathrm{mm}$ apart

☐ ACCUMULATOR MANAGEMENT SYSTEM

► Install TS accumulator, activate LVS, and connect laptop to the AMS
194 ◯ TS accumulator’s AMS data can be displayed
195 ◯ If Li-Ion LV battery: LV battery’s AMS data can be displayed
► Disconnect TS accumulator
● AMS indicator light . . .
196 ◯ . . . is illuminated red
197 ⊙ . . . is inside the cockpit and marked with “AMS”
198 ◯ . . . is visible in bright sunlight, even from outside
199 ◯ . . . is visible for the driver

☐ TS WIRING

200 ⊙ TS wiring channels are orange
201 ⊙ No other wires than TS wires are orange
202 ⊙ TS wiring outside electrical enclosures in separate non-conductive conduit or orange shielded cable
203 ⊙ Securely anchored to withstand at least $200\,\mathrm{N}$, if outside of enclosure
204 ⊙ Located out of the way of possible snagging or damage
205 ⊙ Shielded against rotating/moving parts
206 ⊙ TS and LV wires separated (n/a for interlock)
207 ⊙ Possible to clearly assign and prove gauge, temperature, and voltage rating of TS wires
208 ⊙ All TS wires and connectors have proper overcurrent protection
209 ⊙ Suitable temperature rating for used position
210 ⊙ Positive locking mechanism on every screwed connection. (Photographs for all inaccessible TS connections)
211 ⊙ TSMPs: positive locking mechanism on every connection. (Photographs for all inaccessible TS connections)
212 ◯ Insulation is not insulating tape or rubber-like paint
213 ⊙ All energy from accumulator flows through the data logger
214 ⊙ TS wires of outboard wheel motors must not be able to reach the cockpit opening in case of a wire break. The wiring outside of the impact structure is the shortest possible distance.
▶ Check interlocks on ...
215 ⊙ ... TS accumulator container(s)
216 ⊙ ... Inverters
217 ⊙ ... Power distribution boxes
218 ⊙ ... Data logger enclosure
219 ⊙ ... Outboard wheel motors have a dedicated interlock wire routed along the TS wiring, must act before the TS wiring or its clamping fails
220 ⊙ ... Outboard wheel motors have a dedicated interlock wire routed along a suspension member, must act if the suspension fails
221 ⊙ Interlocks can be opened for demonstration

☐ TS ENCLOSURES

▶ Check for each TS enclosures ...
222 ⊙ ... Not possible to reach any TS potentials with insulated test probe ($100\,\mathrm{mm}$ length, $6\,\mathrm{mm}$ diameter)
223 ⊙ ... Protected from moisture
224 ⊙ ... HV warning stickers placed (triangle with a black lightning bolt on yellow background)
225 ◯ Data logger is fully enclosed in a housing
226 ◯ Data logger is rigidly mounted by only using the two preapplied 3M™ Dual Lock™ strips on the bottom
227 △ Check data logger functionality and connectivity
228 ◯ Seal data logger enclosure

☐ GROUNDING CHECKS

- A conductive part is grounded when having $\leq 100\,\mathrm{m\Omega}$ measured at $1\,\mathrm{A}$ to LVS ground and being able to continuously carry $\geq 10\,\%$ of the TS main fuse.
- It is possible to join two TS enclosures, one following EV 3.1.1 point 1 and the other one following point 2, if each individual is fully closed.

▶ Check for each TS enclosure ...
229 ◯ ... all materials used to build a TS enclosure separately have a resistance $\geq 2\,\mathrm{M\Omega}$ @ $500\,\mathrm{V}$ ⇒ fully isolated TS enclose, no grounded layer needed
230 ◯ ... expect e.g. screws, (shielded) connectors, backing plates isolating materials used ⇒ fully isolated TS enclose, no grounded layer needed but protruding elements must be properly grounded
231 ◯ ... at least one material has $< 2\,\mathrm{M\Omega}$ ⇒ $\geq 0.5\,\mathrm{mm}$ thick solid grounded layer made of aluminium or better (TSAC: $\geq 0.9\,\mathrm{mm}$ thick steal layer) required and properly grounded
▶ Measure resistance of conductive parts to LVS ground (max. $100\,\mathrm{m\Omega}$ @ $1\,\mathrm{A}$) ...
▶ Use second LVS ground measurement point for 4-wire connection if available.
232 ◯ ... main hoop
233 ◯ ... seat mounting points
234 ◯ ... driver harness mounting points
235 ◯ ... firewall mounting points
236 ◯ ... TS firewall
237 ◯ ... TS accumulator container
238 ◯ ... TS motors
239 ◯ ... TS enclosures if applicable
240 ◯ ... TS enclosure protruding parts if applicable
241 ◯ Each grounding is able to carry $\geq 10\,\%$ of TS main fuse
242 △ Conductive parts within $10\,\mathrm{cm}$ around TS part are $\leq 100\,\Omega$ to LVS ground

☐ DIS-CHARGE CIRCUIT AND BODY PROTECTION RESISTORS

▶ Switch off LV. Measure resistance between TS+ and TS- measuring points
243 ◯ Resistance is $30\,\mathrm{k\Omega}$ [9] + discharge resistor

☐ INSULATION MEASUREMENT TEST

▶ Choose test voltage to $500\,\mathrm{V}$ [10]
▶ Connect insulation tester to TS+ and GND
▶ Measure resistance: $R_{\mathrm{iso+}}$ = MΩ
244 ◯ Resistance is much higher than $315\,\mathrm{k\Omega}$ [11]
▶ Connect insulation tester to TS- and GND
▶ Measure resistance: $R_{\mathrm{iso-}}$ = MΩ
245 ◯ Resistance is much higher than $315\,\mathrm{k\Omega}$ [11]
246 ◯ Resistances are nearly equal

[9] 2 x Body Protection Resistor (BPR)
[10] $U_{max} \leq 250\,V_{DC}$ U_{Test} = $250\,V_{DC}$; $U_{max} > 250\,V_{DC}$ U_{Test} = $500 V_{DC}$
[11] Minimal Resistance = $500\,\Omega/\mathrm{V} \cdot V_{\mathrm{max}}$ + BPR

!! TEST AT HIGH VOLTAGE !!

☐ TRACTIVE SYSTEM POWER-UP

- ▶ All driven wheels are off the ground, driven wheels removed
- ▶ Connect multimeter between TS+ and TS-
- ▶ Switch on TSMS with LVMS deactivated

247 ◯ Voltage at TS measurement points $\leq 60\,\mathrm{V\,DC}$

- ▶ Switch on LVMS with TSMS deactivated

248 ◯ IMD and AMS indicator illuminate for $1\,\mathrm{s}$ to $3\,\mathrm{s}$ for visible check

249 ◯ Voltage at TS measurement points $\leq 60\,\mathrm{V\,DC}$

- ▶ Switch on TSMS and all shutdown buttons
- ▶ Reset any IMD or AMS errors

250 ◯ TS still deactivated

- ▶ Activate TS, measure TS voltage during TS power-up. Use the team's multimeter and test leads. Set multimeter into manual range

251 ◯ System is precharged before second AIR closes

- ▶ Switch off TSMS

252 ◯ TS voltage decreases below $60\,\mathrm{V\,DC}$ within $5\,\mathrm{s}$

- ▶ Try to power-up TS with switched off TSMS

253 ◯ TS still deactivated

- ▶ Switch on TSMS

254 ◯ TS still deactivated

☐ TRACTIVE SYSTEM SHUTDOWN

- ▶ Connect multimeter between TS+ and TS-
- ▶ Check first and two more of the following switches, deactivation leads to TS shutdown, the voltage decreases below $60\,\mathrm{V\,DC}$ within $5\,\mathrm{s}$

255 ◯ Cockpit shutdown button

256 ◯ Shutdown button left

257 ◯ Shutdown button right

258 ◯ Inertia switch

259 ◯ Break-over-travel-switch

260 ◯ Interlocks

☐ TRACTIVE SYSTEM ACTIVE LIGHT

261 ◯ Max. $75\,\mathrm{mm}$ below the highest point of the main hoop and within the roll-over protected envelope (including mounting)

262 △ Full illuminated surface visible by a person standing $3\,\mathrm{m}$ away from TSAL ($1.6\,\mathrm{m}$ eye height)

263 △ $\leq 10°$ blocked by main hoop

- ▶ Activate LVS

264 ◯ TSAL is green only

- ● TS off indicator light ...

265 ◯ ... is illuminated green

266 ⊙ ... is inside the cockpit and marked with "TS off"

267 ◯ ... is visible in bright sunlight

268 ◯ ... is visible for the driver

- ▶ Activate TS

269 ◯ TSAL flashes red with frequency $2\,\mathrm{Hz}$ to $5\,\mathrm{Hz}$, and CI is off

270 ◯ TSAL is clearly visible (horizontal position, entire illuminated surface)

- ▶ Deactivate TS, disconnect TSAC state detection circuitry connector if applicable[12], activate LVS and TS

271 ◯ TSAL flashes red and CI is off

- ▶ Deactivate TS, reconnect TSAC state detection, disconnect TS from TSAc, connect power supply $>60\,\mathrm{V\,DC}$ [13] to vehicle TS[14], activate LVS

272 ◯ TSAL is both green and red flashing simultaneously and CI is on

- ▶ Disconnect power supply, reconnect TS to TSAC, remove HVD, override HVD interlock (!! cover TS potentials !!), activate LVS and TS

273 ◯ TSAL and CI is off

- ▶ Reinstall HVD

☐ INSULATION MONITORING DEVICE

274 ◯ One IMD ground line is connected to the accumulator container[15] and one ground line is connected to the main hoop by a separate wired connection

- ▶ $R_{\mathrm{Test}} = 135\,\mathrm{k\Omega}$ [16]
- ▶ Activate TS, connect R_{Test} between TS+ and GND

275 ◯ Shutdown circuits opens within $30\,\mathrm{s}$

276 ◯ TS voltage decreases below $60\,\mathrm{V\,DC}$ within $5\,\mathrm{s}$ after shutdown circuit opens

- ● IMD indicator light ...

277 ◯ ... is illuminated red

278 ⊙ ... is inside the cockpit and marked with "IMD"

279 ◯ ... is visible in bright sunlight, even from outside

280 ◯ ... is visible for the driver

- ▶ Try to activate the TS by the required additional action (EV5.11.2)

281 ◯ Reactivation of TS is not possible

- ▶ Push the reset button which is not accessible to the driver, if any and/or restart LVMS

282 ◯ Reactivation of TS is not possible

- ▶ Remove R_{Test}. Wait for $40\,\mathrm{s}$ until IMD resets status output

283 ◯ Reactivation of TS is not possible

- ▶ Push all reset buttons in the cockpit, if any

284 ◯ Reactivation of TS is not possible

- ▶ Push the IMD reset button which is not accessible to the driver, if any

285 ◯ Reactivation of TS is possible

- ▶ Push *and hold* the reset button which is not accessible to the driver, if any
- ▶ Connect R_{Test} between TS- and GND

286 ◯ Shutdown circuits opens within $30\,\mathrm{s}$

287 ◯ IMD indicator light illuminates

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[12] Skip test if disconnecting the connector also opens the interlock and/or stops LVS supply

[13] $25\,\mathrm{V\,AC\,RMS}$ equal $42.5\,\mathrm{V\,DC}$ when the signal is sinusoidal

[14] Do not use measuring points. The team needs to provide a method of connection that uses the same receptacles as used for TSMP

[15] or the IMD's enclosure

[16] R_{Test} = (max. TS voltage $\cdot\ 250\,\Omega/\mathrm{V}$) - BPR

☐ READY TO DRIVE ACTIVATION SEQUENCE

- ▶ Activate TS, press torque pedal

288 ◯ No turning of motors

- ▶ Let the team set the vehicle to ready-to-drive mode

289 ◯ Pressing brake pedal WHILE activating is necessary

- ▶ Repeat the activation sequence, but push the brake pedal only once before finally pushing the activation button

290 ◯ No ready-to-drive mode possible

- ▶ Disconnect the brake sensor

291 ◯ No ready-to-drive mode possible

292 ◯ Ready to drive sound duration is $1\,\mathrm{s}$ to $3\,\mathrm{s}$ continuously

293 △ Ready to drive sound is min $80\,\mathrm{dBA}$ ($2\,\mathrm{m}$ around the vehicle)

294 ◯ Ready to drive sound is easily recognizable and no animal sound or song part

☐ APPS AND BSPD

- ▶ Set vehicle to ready to drive state
- ▶ Disconnect $\geq 50\,\%$ of APPS
- ▶ Move the accelerator pedal over the entire pedal travel

295 ◯ Motors do not turn

- ▶ Disconnect all APPS
- ▶ Move the accelerator pedal over the entire pedal travel

296 ◯ Motors do not turn

- ▶ Team simulates $5\,\mathrm{kW}$ power (complete BSPD circuitry must be used), press brake representing hard braking ($>0.5\,\mathrm{s}$)

297 ◯ TS shuts down

298 ◯ Reactivation of TS is only possible after $10\,\mathrm{s}$ without implausibility

- ▶ Reactivate TS. Disconnect the current sensor, press brake representing hard braking ($>0.5\,\mathrm{s}$)

299 ◯ TS shuts down

☐ TIS STATUS UPDATE

- ▶ Set online TIS status to *Passed* or *Failed*

NON-COMPLIANCE / COMMENTS

Item #	Non-compliance / Comment	Inspector Name

APPROVAL

	Inspector Names	Date, Time	Signatures when passed
1.	_____/ _____	_____	_____
2.	_____/ _____	_____	_____
3.	_____/ _____	_____	_____
4.	_____/ _____	_____	_____

PART VI: MECHANICAL INSPECTION

The time limit for this part of the inspection is 75 minutes. Continuation of the inspection is possible after requeueing.
During technical inspection all work carried out on the vehicle must be approved by a technical inspector.

☐ TIS STATUS UPDATE (M-INSPECTION)

- ► Set online TIS status (M-Inspection) to *Present*
- ► Write down inspector names legibly, sign only when passed

☐ COMMENTS

- ► Check comments from first page

☐ VEHICLE WITH TALLEST DRIVER READY TO RACE

300 ◯ **PUSH BAR (red color)** - Securely attached to vehicle, detachable, push & pull function for 2 people. University must be written on. Two pair of HV gloves in protecting case and multimeter must be installed. The inspection sheet must always stay with the push bar.

301 △ **CAMERAS** - Secured by two points, see T13.5. No helmet-mounted cameras.

302 △ **VISIBILITY** - Minimum 100° field of view each side. Head rotation or mirrors allowed; mirrors firmly installed and adjusted.

303 △ **VEHICLE CONTROLS** - All controls, including shifter, must be inside cockpit. No arms or elbows outside the SIS plane.

304 ◯ **DRIVER FLUID PROTECTION** - Rigidly mounted firewall (or heat resistant cover for cooling systems using plain water (except wheel motors and their cooling hoses)). For points less than 100 mm above the bottom of the tallest driver's helmet, the firewall must extend sufficiently far upwards and/or rearwards to prevent a direct line of sight to: cooling system and low voltage battery.

305 ◯ **ROLL BAR PADDING** - Must cover areas that could be hit by the driver's helmet with 12 mm thick, SFI spec 45.1 or FIA 8857-2001 padding.

306 △ **OTHER SIDE TUBES** - Design must prevent driver's neck from hitting bracing or side tubes.

307 ◯ **HEAD RESTRAINT**- Near vertical. Must take 890 N load. 40 mm thick, SFI 45.2 standard. Max. 25 mm from helmet. Helmet contact point 50 mm min. from any edge. May be changed for different drivers. Minimum 150x150 mm.

308 ◯ **DRIVER RESTRAINT HARNESS** - SFI 16.1, SFI 16.5, SFI 16.6, FIA 8853/2016. 6- or 7-point system – Two-piece lap belt (min. width 50 mm), two shoulder straps (min. width 75 mm) and two leg or anti-submarine straps (min. width 50 mm). (7-point system must have three anti-submarine straps). Must be securely attached to prim. structure (25.4 x 2.4 mm or equal.).

309 ◯ **LAP BELT MOUNTING** - Pivoting mounting with eye bolts or shoulder bolts attached securely to primary structure. Min. tab thickness 1.6 mm. Attachment brackets to the monocoque must be steel, see T4.5.5.

310 ◯ **LAP BELT POSITIONING** - Must pass over pelvic area between 45 - 65 deg. to horizontal for upright driver, 60-80 deg. for reclined. The lap belts must not be routed over the sides of the seat.

311 ◯ **SHOULDER HARNESS MOUNTING** - Mounting points 180 - 230 mm apart (measured center to center). Attached to primary structure - 25.4 x 2.4 mm or 25.0 x 2.5 mm steel tube min. NOT to put bending loads into main hoop bracing without extra bracing. Additional braces if not straight to main hoop. Cannot pass through a firewall. Attachment brackets to the monocoque must be steel.

312 △ **SUSPENSION** - Fully operational with dampers front and rear; 50 mm minimum wheel travel (minimum jounce of 25 mm) with driver in vehicle.

☐ VEHICLE WITHOUT DRIVER

313 △ **TECH STICKER SPACE** - $45\,\mathrm{mm}$ x $175\,\mathrm{mm}$ on centerline of front of vehicle in front of the cockpit opening

314 △ **SCHOOL NAME & OTHER DECALS** - School name, or recognized initials - min. 50 mm tall (all letters). on both sides in roman letters. Must be clearly visible.

315 △ **VEHICLE NUMBERS** - On front & both sides of vehicle, minimum 150 mm tall, 20 mm stroke & spacing, 25 mm min. between number and background edge, black on white, white on black only, specified background shapes. Must be clearly visible, font: Roman Sans-Serif characters.

316 △ **BODYWORK EDGES** - edges that could contact a pedestrian must have a minimum radius of 1.0 mm (safety requirement).

317 △ **BODY & STYLING** - Open wheeled, open cockpit, formula style body. Vertical keepout zones 75 mm in front and behind tires (no aero exceptions), tires unobstructed from sides.

318 ◯ **BODYWORK** - Min. 38 mm radius on nose. No large openings in bodywork into driver compartment in front of or alongside driver, (except cockpit opening). In any side view in front of the cockpit opening no external concave radii (exception T8.2).

319 ◯ **AERODYNAMIC DEVICES** - Securely mounted. The deflection may not exceed 10 mm when a force of 200 N is applied over a surface of 225 cm^2 and not more than 25 mm when a point force of 50 N is applied.

320 △ **AERODYNAMICS** - ALL aerodynamic devices maximum 250 mm rearward of rear tires, maximum 700 mm forward of front tires. Devices lower than 500 mm from the ground rearward of the front axle must be no wider than vertical plane from the outside of the front and rear tires. Devices higher than 500 mm behind the front axle must not be wider than the inside of the rear tires.

321 △ **AERO VERTICAL HEIGHT** - Devices forward of a vertical plane through the rearmost portion of the front face of the driver head restraint support, excluding any padding, set to its most rearward position, must be lower than 500 mm from the ground. Rear device max 1.2 m above ground (incl. end plates); Front device max 250 mm above ground outside of the inside plane of the front tires inside this plane max 500 mm.

322 ◯ **EDGES/RADII** - Edges that could contact a pedestrian must have a minimum radius of: forward facing edges min 3 mm; all other edges min. 1 mm.

323 △ **SEAT** - Insulated against heat conduction, convection and radiation. Lowest point no lower than top of of the upper surface of the lowest SIS member OR must have longitudinal, 25.4 x 1.65 mm steel tube underneath.

324 ⊙ **COCKPIT OPENING** - Fig. 11 (left) template passes down from above cockpit to below the upper side impact member. Steering wheel, seat & padding can be removed. No removing of firewall.

325 ◯ **COCKPIT INTERNAL CROSS SECTION** - Fig. 11 (right) template passes from the cockpit opening to 100 mm rear of rearmost pedal contact area (in most forward position). Steering wheel and paddings can be removed (without tools).

326 △ **STEERING WHEEL** - Continuous perimeter, near round (no concave sections) with driver operable quick disconnect. 250 mm max from front hoop.

☐ REMOVE BODY PANELS

327 ◯ **JACKS** - Up to two devices that lift up all driven wheels min. 100 mm above the ground. In lifted position it is safe to enter and exit the vehicle and the devices must not extend out of the vehicles projected surface area. University name must be written on. Vehicle pickup points must be indicated by orange triangles.

328 ◯ **DRIVER'S LEG PROTECTION** - Covers inside of cockpit over any sharp edges or moving suspension / steering components.

329 ⊙ **DRIVER'S FOOT PROTECTION** - Feet must be rearward of the front bulkhead. The front bulkhead, together with the AIP, must cover the driver's feet in front view. No part of shoes or legs above or outside the primary structure (25x1.2 or equivalent) in side or front views when touching the pedals.

330 ⊙ **PERCY** - Helmet of 95th percentile male (PERCY) including 50 mm clearance must be below the lines between top of front and main roll hoops and between top of main hoop to rear attachment point of main hoop bracing. Center of bottom circle placed minimum 915 mm from pedals.

331 ◯ **BRAKES** - Dual hydraulic system & reservoirs, operating on all four wheels, (one brake on limited slip differential is OK). System must be protected by structure or shields from drivetrain failure or minor collisions. No plastic brake lines. No brake-by-wire. No parts below chassis in side view. Brake pedal capable of 2000 N, no failures if official exerts max force (seated normally in vehicle).

332 △ **BRAKE OVER TRAVEL SWITCH** - In the event of a failure in one or both of the brake circuits the brake pedal over travel will result in the shutdown circuit being opened.

333 ◯ **WHEELS** - 203.2 mm (8") min. diam. No aluminium or hollow wheel bolts. Single retaining nut must incorporate a device to retain the nut. Aluminum wheel nuts must be hard anodized.

334 ◯ **RIM CLEARANCE** - The radial clearance between any non-rotating part and the inside of the rim must be at least 5 mm in static condition at any steering angle and any ride height.

335 ◯ **FIREWALL** - Fire resistant material; must separate driver compartment from cooling, oil system & LV battery. Pass-throughs OK with grommets. Multiple panels OK if gaps sealed. No gaps at sides or bottom. Must be rigidly mounted to the chassis. Material must meet UL94-V0, FAR25 or equivalent. On tractive side min. 0.5 mm aluminium plate grounded, on the driver side a rigid insulating layer (no CFRP) UL94-V0 or equivalent should be installed that can withstand a 250 N 4 mm screwdriver penetrating test.

☐ SES, IAD & REQUIRED TESTS PRESENTED

336 ◯ **SES TUBING & MATERIALS** - Team must show an APPROVED SES. No magnesium tubes in primary structure.

337 ⊙ **SES TEST SPECIMEN** - Team must show all relevant test specimen. Labled (non-removable) with structure acronym and date. Speciment width, skin & core thickness according to SES.

338 ⊙ **INSPECTION HOLES** - 4.5 mm inspection holes required in non-critical areas of front & main hoops. Must be accessible with standard calliper. Inspectors may ask for holes in other tubes and/or structures.

339 ⊙ **SES DIMENSIONS & THICKNESSES** - All chassis dimensions according to SES: tube diameter and wall thickness; laminate skin thickness, core thickness, panel height.

340 ◯ **HOLES & CUTOUTS** - All holes/cutouts in primary structure < 60 mm² or deducted from panel height.

341 ⊙ **LAMINATE ORIENTATION** - Tested structures must be correctly oriented or quasi-isotropic (T3.5.4, especially MHBS).

342 ⊙ **BOLTED JOINTS/ATTACHMENTS** in primary structure - Distance hole centerline to the nearest free edge > 1.5 x hole diameter. Steel backing plates (≥2 mm thick) with perimeter near circular or oval used at attachment points (must be fully supported). According to SES (T3.15.6) if two panels are bolted together.

343 ⊙ **HARNESS ATTACHMENTS** for shoulder harness, lap belt and anti-submarine belt according to SES calculation, simulation and/or physical test. Test/calculation conducted according to realistic belt angle.

344 ⊙ **MAIN HOOP** - Must be made of one piece and extend to lowest frame member. Above major structure, must be within 10 deg. of vertical plane. Smooth bends without wrinkles, not oval after bending.

345 ⊙ **MAIN HOOP BRACING** - Same material as main hoop (both (non) magnetic). One straight brace on each side. Attached within 160 mm from the top. Min. 30 deg. included angle with main hoop. No bends. No rod-ends. Proper design for removable braces (capping etc.) on both ends.

346 ⊙ **FRONT HOOP** - Must be closed section metal tube. Can be multi-piece with gussets or additional attachments to the monocoque. Must extend down to lowest frame member. No lower than top of steering wheel. Max. 20 deg. to vertical.

347 ⊙ **FRONT HOOP BRACING** - Two straight forward facing braces, attached within 50 mm of top. Extra rearward bracing required if front hoop leans backwards more than 10 deg.

348 ⊙ **FRONT BULKHEAD SUPPORT** - Upper tube connecting within 50 mm of top of bulkhead, and connecting within 100 mm above and 50 mm below upper SIS tube.

349 ⊙ **SIDE IMPACT PROTECTION** - Upper tube between 240 - 320 mm above lowest inside chassis point between FH and MH.

350 ◯ **SUSPENSION PICK-UP POINTS** - Inspected thoroughly for integrity. No crushed core, no skin detacted from core.

351 ⊙ **FRONT IMPACT PROTECTION** - Team must show an APPROVED IAD and test piece (if applicable), which both must reflect status on the car. IMPACT ATTENUATOR forward of bulkhead. IA must be securely fastened directly to AIP capable of taking transverse & vertical loads (no tape, etc.). Aerodynamic devices and/or sensors in front of the front bulkhead must follow T3.18.

352 ⊙ **IAD DIMENSIONS** - IA min. 200 mm long x 200 mm wide x 100 mm high. AIP solid sheet metal, min. 1.5 mm steel or 4.0 mm aluminium; alternative design accepted, if SES/IAD approved. Standard IA: Requires diagonal or X-brace if FBH dimensions larger than 400 mm width and/or 350 mm height.

353 △ **IA POSITION** - The minimum volume dimensions cannot not be more than 350 mm above ground (measured with driver seated).

354 ⊙ **AIP ATTACHMENT** - Standard: must be welded (full perimeter, size: min. to centerlines) or min. 8 screws M8 grade 8.8 (critical fasteners T10) (size: min. outside dimensions). Non-standard: Must follow T3.15.6.

☐ VEHICLE LIFTED AND WHEELS REMOVED

355 ◯ **FASTENERS** - Steering, braking, harness and suspension systems must use SAE grade 5 or metric grade M8.8 or higher specs (AN/MS) with visible positive locking mechanisms, no adhesive or lock washers. Minimum of 2 exposed threads with lock nuts. Rod ends in single shear are captured by a washer larger than the ball diameter. Adjustable tie-rod ends must have jam nuts to prevent loosening. No nylon lock nuts for brake calipers or brake discs. No button head cap, pan head or round head screws in critical locations, e.g cage structure or harness mount. Primary structure e/D > 1.5. Snap or retaining rings must not bear any load in non-OEM application (e.g. not for brake disc floaters). Adjustable pedals require at least a positive stop with positive locking.

356 ◯ **STEERING** - All steerable wheels must have positive stops

placed on the rack to prevent linkage lock up or tires from contacting any part of the vehicle. Stationary parts within rollover protection envelope. 7 degrees max. free play at the steering wheel. NO STEER-BY-WIRE on front wheels. Rear wheel steering, max. 6 deg. and mechanical stops installed. Bonded joints in accordance with T3.2.8.

357 △ **FLOOR CLOSEOUT PANEL** - Required from foot area to firewall; solid, non-brittle material; multiple panels are OK if gaps less than 3 mm.

358 ○ **GAS CYLINDERS LOCATION** - Axis not pointed at driver, within the rollover protection envelope (see FIGURE 2), insulated from any heat source, must be shielded from the driver. The shields must be steel or aluminum with a minimum thickness of 1 mm.

359 ○ **GAS CYLINDERS** - Proprietary manufacture & labeled, non-flammable gas, regulator on tank, securely mounted, appropriate lines & fittings. Positively retained, i.e. no tie-wraps. Maximum of 10 bar allowed, except cylinders/tanks with directly mounted pressure regulator (-> 10 bar).

360 ○ **SCATTERSHIELDS INCL. MOUNTING** - Required for clutches, chains, belts, etc. No holes. 6 mm diam. grade 8.8 minimum. End parallel to lowest part of the sprocket/pulley in front and rear.

361 △ **SCATTERSHIELD MATERIALS** - For chains, 2 mm min. thick solid STEEL, 3 x chain width. For belts, 3 mm min. thick Al 6061-T6, 3 x belt width. Finger guards: cover all drivetrain parts that spin while vehicle is stationary. No holes >12 mm dia.

362 ○ **LV BATTERY** - Attached securely to frame or chassis.

363 ○ **HIGH PRESS HYDRAULICS** - Pumps and lines must have 1 mm steel or aluminium shields protecting driver and workers.

364 △ **COOLANT** - 100% water. NO ADDITIVES WHATSOEVER or oil for electric motors.

365 ○ **CATCH TANKS** - Any coolant overflow or lube system vents must have separate catch tanks. 0.9 l minimum each, 100 deg. C material, behind firewall, below shoulder level. 3 mm min. dia. vent away from driver down to the bottom level of frame. Cooling systems using plain water, unless sealed, require 100 ml catch tanks.

366 △ **FLUID LEAKS** - Oil, grease, coolant,Brake fluid -> none permitted

367 ⊙ **BELLYPANS** - In total minimum of two venting holes of at least 25 mm diameter in the lowest part of the structure to prevent accumulation of liquids. One in each enclosed chassis structure. Additional holes are required when multiple local lowest parts exist in the structure.

368 ○ **ROTATING PARTS** - Finger guards are required to cover any parts (e.g. fans) that spin while the vehicle is stationary. No holes >12 mm dia.

369 ○ **FANS** - The total rated power of all active devices designed to move air is max. 500W in total.

370 ⊙ **ACCUMULATOR CONTAINER POSITION** - All accumulator containers must lie within the primary structure of the frame lower than the top of the SIS. All accumulator containers must be protected from side, rear and front impact collisions (EV5.5.1, EV5.5.2). If an accumulator container or parts of it are mounted outside of the primary structure an additional impact structure according to T3.14 must be build to protect the accumulator.

371 ⊙ **ACCUMULATOR CONTAINER ATTACHMENT** - Accumulator container must be attached to the primary structure with fasteners min. grade 8.8. Fasteners have to follow T10. Mounting as designed in SES. Brackets 1.6 mm steel or 4 mm aluminium with gussets to withstand bending loads. Monocoque needs 2 mm steel backing plates with perimeter near circular or oval. Equivalent attachment may be according to SES.

372 ⊙ **POSITION OF TRACTIVE SYSTEM PARTS** - All parts belonging to the tractive system must be located within the rollover protection envelope, excluding outboard motors (EV4.4.3).

373 ⊙ **PROTECTION OF TRACTIVE SYSTEM PARTS** - If tractive system parts are mounted in a position where damage could occur from a rear or side impact (below 350 mm from the ground), they have to be protected by a fully triangulated structure with tubes of a minimum outer diameter of 25.4 mm and a minimum wall thickness of 1.25 mm or equivalent.

374 ⊙ **MOTOR CASING** - Min. 2 mm aluminium 6061-T6. May be split into two equal sections. If motor casing is rotating around the stator or is perforated an additional 1 mm aluminium 6061-T6 scatter shield around the motor should be installed.

☐ TIS STATUS UPDATE (M-INSPECTION)

► Set online TIS status (M-Inspection) to *Passed* or *Failed*

TILT TEST

☐ TIS STATUS UPDATE

► Set online TIS status to *Present*

► Write down inspector names legibly, sign only when passed

☐ COMMENTS

► Check comments from first page

☐ TILT TEST

○ **FLUID LEAKAGE** - No fluid spill permitted when vehicle is tilted to 60 degrees in the direction most likely to create spillage. Tanks must be filled to scribe line.

○ **VEHICLE STABILITY** - All wheels in contact with tilt table when tilted to 60 degrees to the horizontal.

403 △ **GROUND CLEARANCE** - At least 30 mm min. with driver. Active suspension in lowest position.

☐ TIS STATUS UPDATE

► Set online TIS status to *Passed* or *Failed*

NON-COMPLIANCE / COMMENTS

NON-COMPLIANCE / COMMENTS

Item #	Non-compliance / Comment	Inspector Name

APPROVAL

	Inspector Names	Date, Time	Signatures when passed
1.	_____/ _____	_____	_____
2.	_____/ _____	_____	_____

PART VII: RAIN TEST

☐ TIS STATUS UPDATE

- ► Set online TIS status to *Present*
- ► Write down inspector names legibly, sign only when passed

☐ COMMENTS

- ► Check comments from first page

☐ RAIN TEST

- ► Apply seal sticker to all additional sealing material, that can be removed (e.g. tape, as not mentioned in IN1.5.1).
- ► The vehicle is lifted off the ground. Tractive system has to be active (TSAL ON)

375 ◯ Tractive system voltage is present at TSMPs

- ► **RAIN PROOF** - No driver is allowed to sit in the vehicle during the test. Water like rain will be sprayed at the vehicle for 120 sec. Another 120 sec. of waitng without water spary.

376 ◯ The Insulation Monitoring Device does not react and not shut down the tractive system.

- ► Connect R_{Test} between any TSMP and LVS GND.

377 ◯ Shutdown circuits opens within 30 s.

☐ TIS STATUS UPDATE

- ► Set online TIS status to *Passed* or *Failed*

NON-COMPLIANCE / COMMENTS

Item #	Non-compliance / Comment	Inspector Name

APPROVAL

	Inspector Names	Date, Time	Signatures when passed
1.	_____ / _____	_____	_____

PART VIII: BRAKE TEST

☐ TIS STATUS UPDATE

- ► Set online TIS status to *Present*
- ► Write down inspector names legibly, sign only when passed

☐ COMMENTS

- ► Check comments from first page

☐ BRAKE TEST

378 ◯ **BRAKING PERFORMANCE** - Must lock all four wheels and stop the vehicle in a straight line at the end of an acceleration run specified by the officials without electrical braking from motors. The tractive system has to be shut down by the driver before braking. The Tractive System Active Light has to be Green during breaking or shortly after the vehicle stopped (may take up to 5 sec. after shut down).

379 ◯ **BRAKE LIGHT** - has to be clearly visible even in bright sunlight.

380 ◯ **DRIVABILITY** - Vehicle must be able to continue driving after braking under its own power without external assistance.

- ► Check brake calipers, brake pedal and connections for leaks.

☐ TIS STATUS UPDATE

- ► Set online TIS status to *Passed* or *Failed*

NON-COMPLIANCE / COMMENTS

Item #	Non-compliance / Comment	Inspector Name

APPROVAL

	Inspector Names	Date, Time	Signatures when passed
1.	_____ / _____	_____	_____